1 • Shared evidence: circuit fault clinic

Circuit fault clinic

Name ______________________ Date ____________

All observations on this page are constructed sample observations from a 1.5 V cell and matching lamp. No real appliance is tested.

Case A • Copper

Bare copper lights the lamp. Copper control works before and after. Clips have sound contact.

Case B • Dry plastic

No visible glow. Copper control lights before and after. Clips touch the dry plastic correctly.

Case C • Foil

No visible glow. The copper control also gives no glow after the test.

Case D • Coated metal

No visible glow. Copper controls work, but both clips touch paint rather than the metal core.

Case E • Aluminium

Bare aluminium lights the lamp. Copper controls work before and after; contact is sound.

W1. Sort A-E into conductor evidence, insulator evidence in this test, or uncertain. Explain two decisions.

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W2. For C and D, what should the teacher check before a material claim?

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Additional constructed observations for the response routes: dry card has no visible glow with valid contacts and passing copper checks. Copper and bare aluminium light; rigid dry plastic does not visibly light. These observations do not measure exact conductivity.

2 • Supported route: build a sound claim

Use page 1. Point, draw or dictate if useful. Use the same scientific goal.

S1. List or draw the cell, lamp and sample in one connected loop. Does a gap in a wire help the lamp light?

Case	Lights / no glow	Conductor / insulator evidence / uncertain
A: copper		
B: plastic		
C: foil		

S2. Complete the three rows using the case evidence.

S3. Complete: Case C is uncertain because the ____ also failed. The teacher should check the ____ before testing again.

S4. Choose copper or plastic for each job. Inside path: ____ because ____ . Outer cover: ____ because ____ .

S5. Does no visible glow prove exactly zero current, or only that this lamp did not visibly light? Explain or point.

3 • Standard route: a diagnostic report

Use all cases on page 1. A sound report separates valid evidence from an uncertain result.

Sample/case	Observed lamp	Classification	Reason
Copper / A			
Plastic / B			
Foil / C			
Coated metal / D			
Aluminium / E			

N1. Complete the observation and classification table. Refer to controls or contact where relevant.

N2. Write a safe three-step checking sequence before classifying an unlit result.

3 continued • Standard route: the material decision

Continue the same route. Use the common evidence page.

N3. Draw a cable cross-section. Label a suitable inner conductor and outer insulator. Explain both material choices.

N4. Why does a dark lamp not prove a material is safe at mains voltage?

N5. Can these observations tell you whether copper conducts exactly twice as well as aluminium? Explain.

4 • Stretch route: when the indicator misses a current

Use page 1. This is supplied evidence, not an instruction to change the classroom circuit.

Constructed evidence from a separate, teacher-controlled measuring circuit: a sensitive current meter detects a small current through graphite. The same lamp gives no visible glow. Checks before and after work. Graphite is a form of carbon and is not a metal.

T1. Evaluate: “Only metals can conduct electricity.” Use the graphite evidence.

T2. Explain why a lamp can miss a conducting sample. What does no visible glow actually tell you?

4 continued • Stretch route: a stronger test

Continue the same route. Use the common evidence page.

T3. Design a fair comparison of two prepared solid samples with the approved lamp circuit. Include the control checks, contact check and a limitation of this method.

T4. Use cases C and D to explain two different reasons why an unlit lamp may lead to an uncertain classification.

T5. Justify an inner conducting path and an insulating outer layer for a low-voltage lead. Include one limit of what our test can certify.

5 • Exit and investigation record

Record your own reasoning. If you performed a practical, label observations as your results; otherwise write paper evidence route.

E1. Why is a connected conducting loop needed in our lamp circuit?

E2. What do the known-copper checks tell you?

E3. Name a useful inside and outside material for a lead and explain their jobs.

Optional actual test / paper route	Prediction	Observation	Control passed?
Sample 1:			
Sample 2:			
Sample 3:			

E4. What would you improve next time to make a diagnosis more trustworthy?

Reflection. A claim I changed or qualified was …